

Question number	Answer	Mark
3(a)(i)	The correct answer is B <i>A is incorrect because individual 1 is not homozygous</i> <i>C is incorrect because genotype and phenotype are the wrong way round</i> <i>D is incorrect because genotype and phenotype are the wrong way round</i>	(1)

Question number	Answer	Mark
3(a)(ii)	The correct answer is D <i>A is incorrect because individual 3 is not affected but his parents are both carriers</i> <i>B is incorrect because individual 3 is not affected but his parents are both carriers</i> <i>C is incorrect because individual 3 is not affected but his parents are both carriers</i>	(1)

Question number	Answer	Mark
3(a)(iii)	The correct answer is C <i>A is incorrect because 1 in 4 CMT × 1 in 2 female = 1 in 8</i> <i>B is incorrect because 1 in 4 CMT × 1 in 2 female = 1 in 8</i> <i>D is incorrect because 1 in 4 CMT × 1 in 2 female = 1 in 8</i>	(1)

Question number	Answer	Additional guidance	Mark
3(b)(i)	6 300 / 6.3×10^3 (1)	ACCEPT $6.3 \cdot 10^3$	(1)

Question number	Answer	Additional guidance	Mark
3(b)(ii)	An answer that includes the following points: - the {genotype / alleles present in the gametes} of parents shown (1) - possible genotypes of children shown (1) - tt is unaffected and Tt has CMT (1)	(Parent 1) (Parent 2) tt Tt T or t tt (tt) Tt (Tt) Not CMT/ no / x CMT / yes / ✓ ACCEPT from a pedigree diagram that shows the Ts and ts NB ECF if genotypes / gametes wrong for mp 1	(3)

Question number	Answer	Additional guidance	Mark
3(c)	An explanation that includes the following points: - male parent with CMT will have {X^c Y / X^T Y/ the CMT allele on the X chromosome (only) / no CMT allele on the Y chromosome} (1) (only) the Y chromosome is passed onto their sons (1)	ACCEPT mutation (for CMT) on X chromosome CMT version of the gene faulty allele (for CMT) IGNORE CMT gene / CMT / the disease ACCEPT does not pass X chromosome on	(2)